



# Fabrication of bio-based amphiphilic hydrogel coating with excellent antifouling and mechanical properties

Guangming Lu<sup>a,b</sup>, Shu Tian<sup>a,c</sup>, Jingyu Li<sup>a</sup>, Yongjian Xu<sup>b</sup>, Shuan Liu<sup>a</sup>, Jibin Pu<sup>a,\*</sup>

<sup>a</sup> Key Laboratory of Marine Materials and Related Technologies, Zhejiang Key Laboratory of Marine Materials and Protective Technologies, Ningbo Institute of Materials Technology and Engineering, Chinese Academy of Sciences, Ningbo 315201, PR China

<sup>b</sup> School of Marine Sciences, Ningbo University, Ningbo 315820, PR China

<sup>c</sup> Department of Biochemical Engineering, School of Chemical Engineering and Technology, Tianjin University, Tianjin 300072, PR China

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## ABSTRACT

Antifouling coatings are crucial to protect marine facilities and aquacultures from the damage of microorganisms for long-term serves. However, it is still a challenge to fabricate coating with excellent antifouling performance and superior mechanical stability. Here, we report a bio-based amphiphilic hydrogel coating with excellent antifouling and mechanical properties. The coating was achieved by the in-situ formation of hydrophobic and hydrophilic interpenetrating polymer network (IPN). The hydrophobic part from a synthesized silicon-containing epoxy resin contributes to superior mechanical properties including high tensile strength and excellent adhesion of 5B. While the hydrophilic hydrogel part from the cross-linking of a hydrophilic polymer with AgNPs gives excellent antifouling properties, resist to proteins, bacteria, algae, and other marine organisms. The antifouling performance of the coating was evaluated by the attachment of proteins (BSA-FITC), bacteria (*Escherichia coli* and *Bacillus subtilis*), and Algae (*Navicula torquatum* and *Phaeodactylum tricornutum*). The results show that the bio-based amphiphilic hydrogel coating (e.g. SA-1-5) endows surface with excellent resistance to the fouling of proteins, bacteria and microorganisms. The overall antifouling and mechanical properties of the amphiphilic coating was also evaluated by field test in East China Sea from June 3rd 2020 to July 17th 2020 that the bio-based amphiphilic hydrogel coating was intact and almost no marine organisms attached. This work provides a new strategy for the fabrication of bio-based and high-performance antifouling coatings.

## 1. Introduction

Marine biofouling is caused by the undesirable settlement and colonization of marine organisms, such as marine microorganisms, plants and animals on the submerged surfaces,[1–6] which leads to serious economic, environmental and health issues for many application fields. For example, in the shipping industry, biofouling not only increases the weight of the hull, but also causes the rise of frictional resistance. As a result, fuel consumption and greenhouse gas emissions are significantly increased.[7–8] In marine facilities, biofouling initiates and accelerates the corrosion of concrete and metal structures, and thus the cost of maintenance rises significantly.[9–11] In aquaculture, biofouling blocks water exchange through net mesh and deforms the cage due to the increase of extra weight.[12–14]

Antifouling coatings are the most convenient, effective and economic strategy to prevent marine biofouling.[15–20] Self-polishing

copolymer-based coatings containing tributyltin (TBT) have been well established and commercialized for its high efficiency in preventing the formation of the biofouling.[21] However, TBT was globally banned in 2008 because of its toxicity to non-target organisms,[22] and subsequent side-effects on marine ecosystems. Since then, it's urgent to develop highly efficient and eco-friendly antifouling materials.

The fouling release coatings (FRCs), made of silicone or fluorinated polymers and following non-stick mechanism, have become the most popular strategy to prevent fouling on ship hulls.[23–24] The low surface energy and low modulus of such polymers could minimize the attaching ability of biomolecules and organisms.[25] Silicone-based FRCs such as polydimethylsiloxane (PDMS) elastomers generally outperform fluorinated ones and have been mostly used. However, the silicone-based FRCs can be easily detached from the substrate (usually epoxy primer) due to the poor-sticking ability of siloxanes. Various approaches have been developed to enhance the adhesion strength

\* Corresponding author.

E-mail address: [pujibin@nimte.ac.cn](mailto:pujibin@nimte.ac.cn) (J. Pu).

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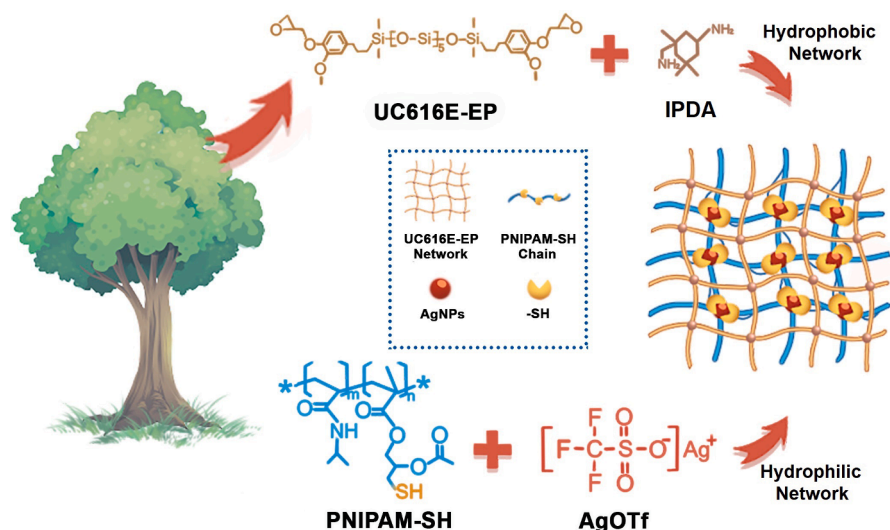
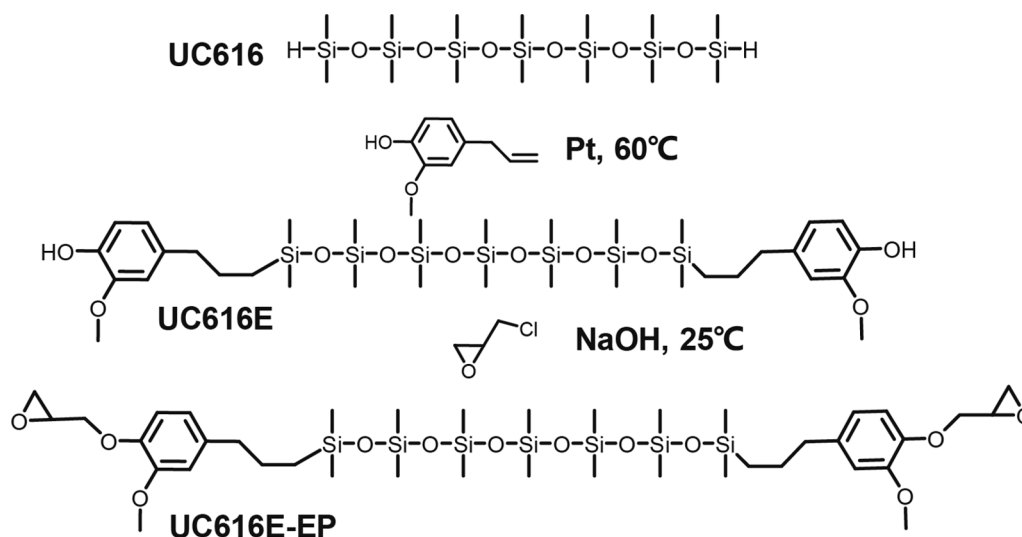


Fig. 1. Schematic illustration for the design of the bio-based amphiphilic hydrogel coating.

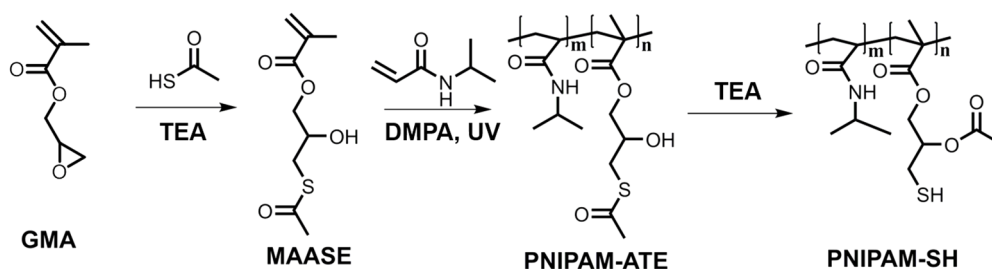
between the silicone-based FRCs and the epoxy primer substrate, such as the introduction of polar interaction, addition of a tie-layer, usage of a coupling agent (e.g., bis-( $\gamma$ -trimethoxysilylpropyl)amine,[26]  $\gamma$ -glycidypropylmethyldiethoxysilane,[27] etc.), and adjusting interface mechanical properties. However, these methods still suffer from high cost, complex processes, poor adhesion and mechanical properties.[28]

As such, numerous strategies including the incorporation of nanofillers[29–31] and silicone-free oils,[32–33] modification with polyurethane[34] and epoxy have been reported to improve the adhesion and mechanical properties of silicone-based FRCs.[35–36] For instance, loading multiwall carbon nanotube and graphene oxide could significantly enhance the elongation at break and tensile strength of PDMS elastomer without affecting its antifouling properties.[29] Although significant advances have been achieved, silicone-based FRCs fail to resist to the attachment of marine slime layers (such as marine bacteria and diatoms).[37–39] To develop FRCs with slime-resistant performance, hydrophilic polymers have been introduced into PDMS to form amphiphilic coatings. The hydrophobic PDMS could make the marine organisms be easily detached from the coatings under hydrodynamic forces, and the hydrophilic polymers could tightly bind water molecules to obtain a hydration layer which could resist marine organism adsorption.[40–44] In our previous work,[45] we developed an

amphiphilic silicone-based antifouling coating from a nanocomposite hydrogel consisting of silver nanoparticles and grafted polymer chains, which shows multiple antifouling characteristics of hydrophilic hydrogel, functional nanoparticles and FRCs. However, the mechanical properties of the coating are still poor, and epoxy primer from non-sustainable fossil resources is required. Developing bio-based antifouling coatings from renewable bioresources could address this resource issue. Recent years have witnessed the rapid development of bio-based materials due to their ecological and economic benefits. Thus, in this work, we develop a new strategy to prepare bio-based hybrid silicone-based antifouling coating with excellent antifouling and mechanical properties. The bio-based epoxy monomer containing siloxane chain was synthesized from bioresource eugenol and further cross-linked by isophoronediamine (IPDA) to produce hydrophobic bio-based epoxy network. Meanwhile, a mercapto hydrophilic polymer (PNIPAM-SH) was also synthesized and mixed with tri-fluoromethanesulfonic acid silver salt to form hydrophilic nanosilver (AgNPs) hydrogel network, which penetrates into the bio-based epoxy network and generates an amphiphilic hydrogel coating with interpenetrating polymer network (IPN) (Fig. 1). The chemical structures of the bio-based epoxy monomer, mercapto hydrophilic polymer as well as the network structure of the coating were characterized in detail.



Scheme 1. Synthetic route of bio-based epoxy monomer containing siloxane chain UC616E-EP.



Scheme 2. Synthetic route of PNIPAM-SH.

Moreover, the mechanical properties, antifouling properties and related mechanism were investigated thoroughly.

## 2. Experimental section

### 2.1. Materials

Triethylamine (TEA), glycidyl methacrylate (GMA), benzoin dimethyl ether (DMPA), thioacetic acid (TA), N-isopropylacrylamide (NIPAM), trifluoromethanesulfonic acid silver salt ( $\text{AgOTf}$ ), eugenol (purity >99%), 1,1,3,3,5,5,7,7,9,9,11,11,13,13-tetradecamethylheptasiloxane (UC616), organic platinum catalyst (purity greater than 99%), tetrabutylammonium bromide (TBAB) and isophoronediamine (IPDA) were purchased from Aladdin (Shanghai, China). Sodium hydroxide, acetone, petroleum ether, 1,4-dioxane, dichloromethane and chloroform were supplied from Sinopharm Chemical Reagent Co., Ltd., China. The deionized water was produced in our lab.

### 2.2. Synthesis of bio-based epoxy monomer containing siloxane chain UC616E-EP

UC616E-EP was synthesized by the reaction of 1,1,3,3,5,5,7,7,9,9,11,11,13,13-tetradecamethylheptasiloxane with eugenol to achieve diphenol UC616E, followed by reacting with epichlorohydrin, as shown in Scheme 1. 32.8 g (0.2 mol) of eugenol, 2.4 g of an organic platinum catalyst (the molar ratio of double bonds to Pt was  $1:6 \times 10^{-5}$ ), and 100 mL of 1,4-dioxane were placed in 500-mL four-neck flask equipped with a magnetic stirrer, a thermometer, a constant pressure-dropping funnel and a condenser pipe, and stirred at room temperature for 30 min. 50.5 g (0.10 mol) of 1,1,3,3,5,5,7,7,9,9,11,11,13,13-tetradecamethylheptasiloxane (UC616) dissolved in 50 mL of 1,4-dioxane was added dropwise into the above system within 2 h, and the system was heated to 60 °C and reacted for 24 h. After that, the reaction system was precipitated by acetone–petroleum ether component solvent for five times to get two transparent liquid layers. After removing the solvents of the lower liquid layer, a light-yellow liquid UC616E was obtained. Then 41.6 g (0.05 mol) of UC616E, 0.7 g of tetrabutylammonium bromide (TBAB), and

92.5 g (1 mol) of epichlorohydrin were placed in 500-mL four-neck flask equipped with a magnetic stirrer, a thermometer, a constant pressure-dropping funnel and a condenser pipe, and stirred at room temperature for 30 min, and reacted at 65 °C for 5 h. After cooling to room temperature, 15 g of 40 wt% NaOH aqueous solution was added dropwise into the reaction system which was then stirred at room temperature for 6 h. A viscous light-yellow liquid product UC616E-EP was obtained after being washed with water for 5 times and extracted with dichloromethane, followed by removing the solvents and excess epichlorohydrin via rotary evaporation.

### 2.3. Synthesis of hydrophilic polymer PNIPAM-SH

The hydrophilic polymer PNIPAM-SH was synthesized according to the literature.[45] 28.4 g of glycidyl methacrylate was dissolved in 100 mL  $\text{CHCl}_3$ , and 1.7 mL of thioacetic acid was added slowly with a syringe. When the reaction system was cooled to 0 °C in an ice bath, 1 mL of triethylamine was added dropwise with a syringe. After being stirred for 30 min, the ice bath was removed and the reaction system reacted at room temperature for 24 h. After reaction, the mixture was extracted with 100 mL of 1 M HCl aqueous solution and 100 mL of  $\text{NaHCO}_3$  saturated aqueous solution for 1 time and 3 times, respectively. After being dried with anhydrous sodium sulfate for 24 h, the extracted solution was evaporated to remove the solvent  $\text{CHCl}_3$ , and an oily yellow liquid MAASE was obtained. Then, 2 g of MAASE, 10 g of NIPAM and 0.12 g of photo initiator DMPA were dissolved in 50 mL of THF, and the reaction was carried out under UV radiation (365 nm, 26-W lamp) and  $\text{N}_2$  atmosphere at room temperature for 24 h. The white solid PNIPAM-ATE was obtained after precipitation with anhydrous dimethyl ether and being dried in a 50 °C vacuum oven for 24 h. Finally, 10 g of PNIPAM-ATE and 50 mL of THF were placed in 100-mL flask, and 1 mL of triethylamine was added slowly, and reacted at room temperature for 24 h. After reaction, the white solid product PNIPAM-SH was obtained after precipitation with anhydrous dimethyl ether and being dried in a 50 °C vacuum oven for 24 h. The synthetic route of PNIPAM-SH is illustrated in Scheme 2.

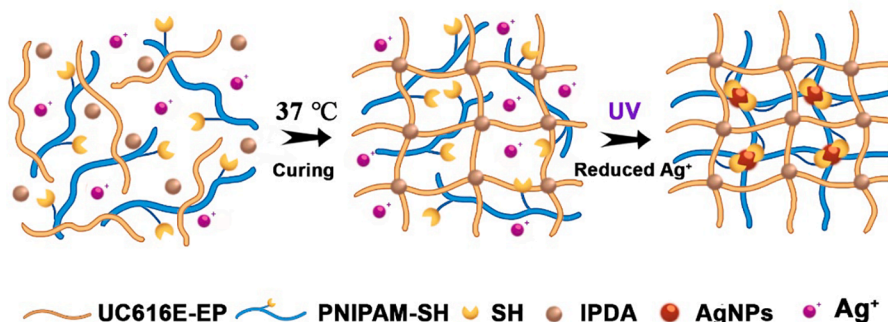


Fig. 2. Preparation process of bio-based amphiphilic hydrogel coating.

**Table 1**

The feed compositions for the coatings.

| Sample | UC616E-EP | IPDA   | AgOTf          | PNIPAM-SH      |
|--------|-----------|--------|----------------|----------------|
| SA-0-0 | 5 g       | 0.45 g | 0              | 0              |
| SA-0-5 | 5 g       | 0.45 g | 0              | 0.27 g (5 wt%) |
| SA-1-0 | 5 g       | 0.45 g | 0.06 g (1 wt%) | 0              |
| SA-1-5 | 5 g       | 0.45 g | 0.06 g (1 wt%) | 0.27 g (5 wt%) |

#### 2.4. Preparation of bio-based amphiphilic hydrogel coating

UC616E-EP and isophorondiamine (1:0.09, w:w) were placed in 300-mL beaker and stirred at room temperature for 30 min to achieve a certain cross-link density. PNIPAM-SH dissolved in anhydrous ethanol was added into the above system and mechanically stirred for approximately 30 min to achieve an even mixture, then AgOTf dissolved in anhydrous ethanol was added and stirred for 30 min to obtain a

homogeneously distributed bio-based epoxy/nanosilver hydrogel hybrid system. The system was degassed in a vacuum oven and coated on glass panels or pretreated epoxy boards, and cured at room temperature to get a bio-based epoxy network via the ring-opening reaction of UC616E-EP with IPDA. The cured samples were radiated under UV lamp for more than 4 h (or sun light for more than 2 days). The appearance of the coatings became from colorless to dull red in virtue of the formation of nanosilver from AgOTf under UV radiation. Then the nanosilver cross-linked PNIPAM-SH through its chelation with thiol group to get nanosilver hydrogel network. As a result, a dull red or black and red bio-based amphiphilic hydrogel coating (SA-1-5) with interpenetrating polymer network (IPN) was obtained. For comparison, coatings without loading PNIPAM-SH and AgOTf (SA-0-0), loading only PNIPAM-SH (SA-0-5) and loading only AgOTf (SA-1-0) were also prepared in a similar method. The preparation process is illustrated in Fig. 2, and the feed compositions for the coatings are shown in Table 1.

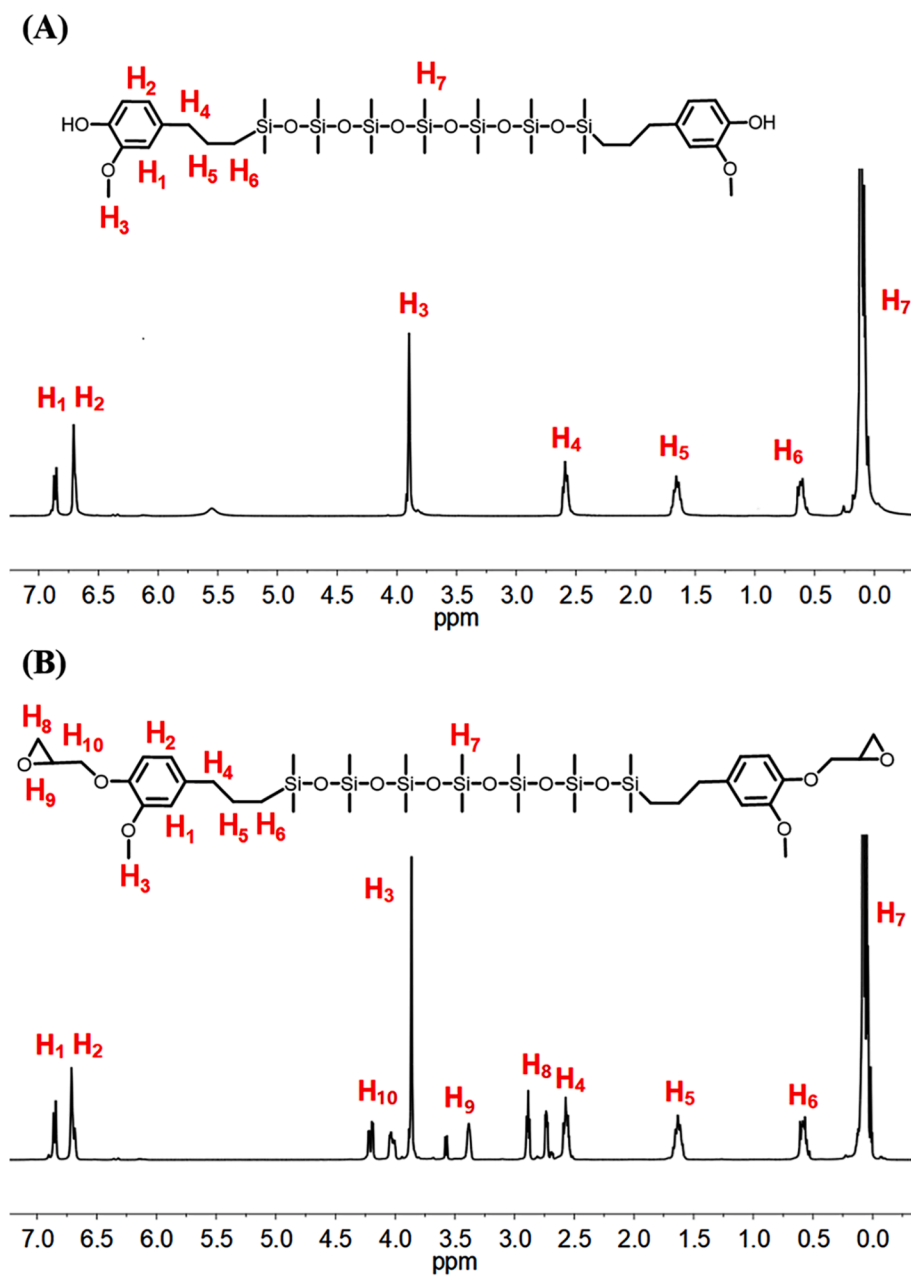


Fig. 3.  $^1\text{H}$  NMR spectra of (A) UC616E and (B) UC616E-EP.

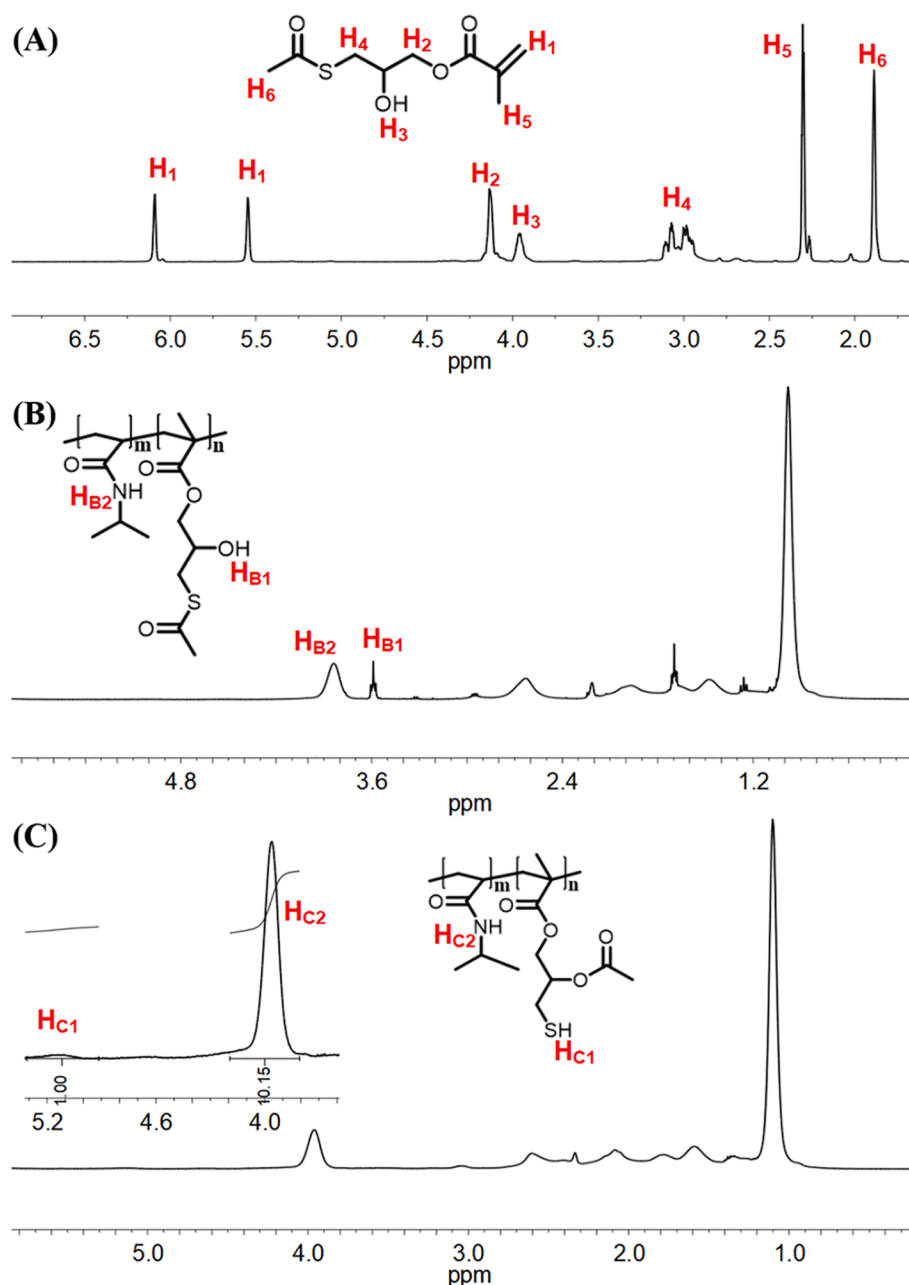


Fig. 4.  $^1\text{H}$  NMR spectra of (A) MASSE, (B) PNIPAM-ATE and (C) PNIPAM-SH.

## 2.5. Characterizations

The size and the morphology of the AgNPs were characterized by Lambda 950 ultraviolet–visible–near infrared absorption spectrum (UV–VIS–NIR) (Perkin-Elmer, USA) and Tecnai F20 Transmission Electron Microscopy (TEM) (FEI, USA), respectively. Due to the transparent or semitransparent appearance, peeled-off coatings were directly examined at wave length of 350–600 nm for UV–VIS–NIR spectra. For the TEM examination, the samples were first packaged by epoxy resin, then cut into ultra-thin slices and put on the mesh copper TEM grids. The diffraction pattern of AgNPs was studied by D8 Advance X-ray diffractometer (Cu  $\text{K}\alpha$  radiation,  $\lambda = 0.154$  nm, BRUKER, GER). The morphology of the coatings was observed by S4800 Scanning Electron Microscope (SEM) (Hitachi, JPN). The glass transition temperatures of the coatings were examined by DSC214 Differential Scanning Calorimeter (DSC) (NETZSCH, GER). The mechanical properties of the free films from the coatings were measured under ambient conditions according to

ASTM 638 using Universal Machine model MTS E42.503 with a cross-head speed of 2 mm/min. 5 parallel specimens were measured and averaged for each sample. The adhesion of the coatings was evaluated according to ASTM D 3359 and ASTM D4541-09.

## 2.6. Protein adsorption tests

BSA-FITC was dissolved in PBS to get a 2 mg/mL solution. The coatings were placed in the BSA-FITC solution and away from light for 4 h. Then the treated coatings were washed with deionized water. Finally, the surfaces of the treated coatings were observed by fluorescent inverted microscope and the obtained images were used to calculate the protein attachment area by software of ImageJ.

## 2.7. Antibacterial tests

*Escherichia coli* and *Bacillus subtilis* were chosen to investigate the

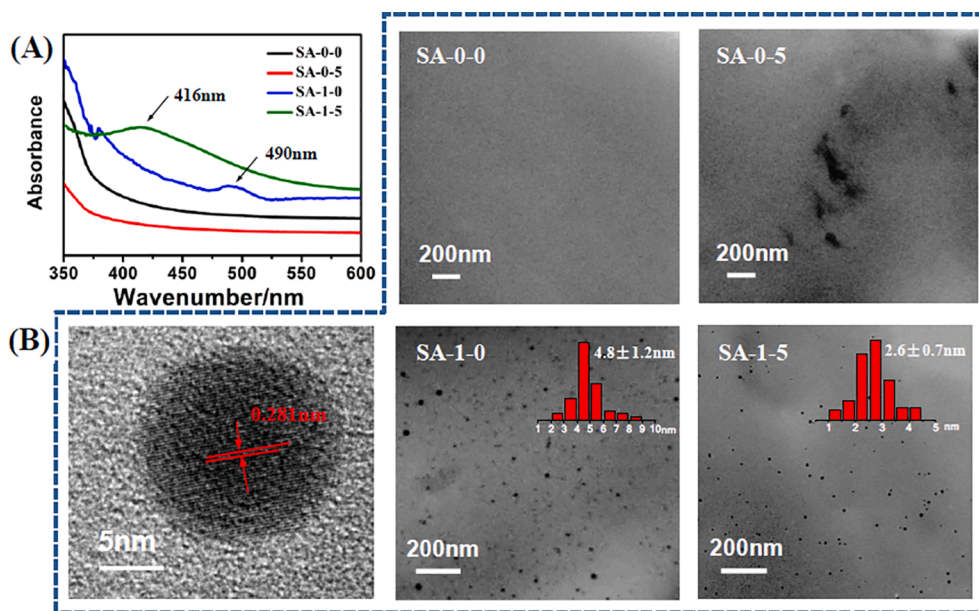


Fig. 5. UV-Vis spectra (a) and TEM images (b) of the coatings.

antibacterial activity of the coatings through shaking flask method. The coatings were added into a sterile 0.9 wt% saline water (40 mL) containing around  $10^5$  CFU/mL of *Escherichia coli* or *Bacillus subtilis* and shaken at 37 °C for 3 h, 6 h, 12 h, or 24 h. Then 0.1 mL of the solution was taken out from the test tube and diluted by ten-fold, and placed on nutrient agar plates in triplicate and incubated at 37 °C for 24 h. The number of bacterial colonies on each plate was counted. The bacterial killing rate ( $R$ ) was calculated by the equation of  $R = (N_c - N_s)/N_c$ .  $N_c$  is the number of microorganism colonies on the control tube (without coating) and  $N_s$  is the number of microorganism colonies on the tube with coating.

*Escherichia coli* was chosen to investigate the bacterial attachment resistance of the coatings. The coatings were placed in 24-well plates and seeded with 100  $\mu$ L of bacterial solution ( $10^6$  CFU/mL). After being cultured at 37 °C for 4 h, the bacteria that were not adhered to the surface of the samples were washed away with PBS, and the samples were stained via Calcein AM/PI kit and observed by fluorescent inverted microscope. The samples were also fixed with glutaraldehyde, washed with ethanol solutions, and observed by SEM after Pt treatment. The obtained images were used to calculate the bacteria attachment by software of ImageJ.

## 2.8. Alga resistance tests

*Navicula torquatum*: NMBguh006-8 and *Phaeodactylum tricornutum*: NMBguh001 (maintained and provided by the Microalgae Collection Center at Ningbo University (Zhejiang, China)) were chosen to investigate the alga attachment on the surface of the coatings. The coatings were placed in 24-well plates and seeded with 1 mL of algae suspension with initial concentration of  $10^8$  cells per mL. After being cultured at 22 °C for 24 h, the algae that were not adhered to the surface of the coatings were washed away with artificial seawater, and the samples were observed by fluorescent inverted microscope, and the obtained images were used to calculate the alga attachment area by software of ImageJ.

## 2.9. Marine field tests

The marine field tests were performed from June 3rd 2020 to July 17th 2020 at the Meishan Island in Ningbo, China (29°49'N, 122°0'E). The samples were coated on stainless steel panels (70 mm  $\times$  100 mm)

and immersed in seawater at a depth of 0.2–2.0 m. After a certain period of time, the coated panels were taken out of the sea, carefully washed with seawater and photographed, and the temperature, oxygen content and pH of seawater were reported.

## 2.10. Contact angle measurement

The water dynamic contact angles were examined by contact angle meter (Dataphysics, GER). A drop of water (5  $\mu$ L) was placed on the surface of the coating at room temperature, and reported the contact angle every 1 min, and the measurement lasted for 10 min.

## 3. Results and discussion

### 3.1. Chemical characterization of bio-based epoxy monomer containing siloxane chain UC616E-EP

Fig. 3 exhibits the  $^1\text{H}$  NMR spectra of UC616E and UC616E-EP. The characteristic peaks of the protons can be identified accordingly. In both  $^1\text{H}$  NMR spectra, the peaks at around 0.1 ppm represent the protons of the Si-CH<sub>3</sub>, the peaks at 6.70 ~ 7.10 ppm belong to the protons on the benzene rings, and the peak at 3.88 ppm corresponds to the protons on the methoxy group. Compared with the  $^1\text{H}$  NMR spectrum of UC616E (Fig. 3A), characteristic peaks at around 2.73 ppm, 2.89 ppm, and 3.38 ppm for epoxy groups and peaks at around 4.05 ppm and 4.21 ppm for methylene close to the epoxy group appear in the  $^1\text{H}$  NMR spectrum of UC616E-EP (Fig. 3B), which indicates that the UC616E-EP was successfully synthesized. The epoxy value UC616E-EP calculated from its molecular weight (945.63 g mol<sup>-1</sup>) is 0.21.

### 3.2. Chemical characterization of PINIAM-SH and its precursors MAASE and PNIPAM-ATE

The chemical structures of MAASE, PNIPAM-ATE and PNIPAM-SH are determined by  $^1\text{H}$  NMR (Fig. 4). As shown in the  $^1\text{H}$  NMR spectrum of MAASE (Fig. 4A), the characteristic peaks for epoxy group disappear, while the peaks for -CH<sub>2</sub>-S- at 3.06 ppm and O = C-CH<sub>3</sub> at 1.89 ppm appear. Other peaks at 6.10 ppm and 5.54 ppm (H, CH<sub>2</sub> = CH-), 4.14 ppm (2H, -CH<sub>2</sub>-O-), 3.95 ppm (1H, -OH), and 2.30 ppm (3H, C = C-CH<sub>3</sub>) match well with the chemical structure of MAASE, which indicates that MAASE was successfully synthesized. As can be seen from

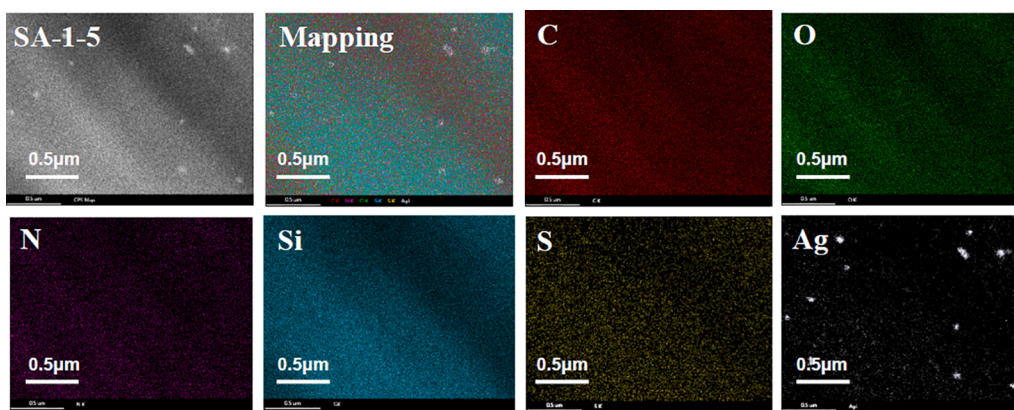


Fig. 6. Elemental mapping images of SA-1-5 by TEM.

the  $^1\text{H}$  NMR spectrum (Fig. 4B), the successful polymerization of MAASE and NIPAM can be demonstrated by the disappearance of peaks of carbon–carbon double bonds, and the peaks at 3.59 ppm and 3.84 ppm of  $-\text{OH}$  ( $\text{H}_{\text{B}1}$ ) and  $-\text{NH}$  ( $\text{H}_{\text{B}2}$ ), respectively. In the  $^1\text{H}$  NMR spectrum of PNIPAM-SH (Fig. 4C), the peak at 3.59 ppm for  $-\text{OH}$  ( $\text{H}_{\text{B}1}$ ) disappears and the peak at 5.13 ppm for  $-\text{SH}$  ( $\text{H}_{\text{C}1}$ ) appears. The peak at 3.96 ppm is ascribed to the proton of  $-\text{NH}$  ( $\text{H}_{\text{C}2}$ ) of PNIPAM-SH. This indicates that the acetyl group was successfully transferred to the  $-\text{OH}$  and released  $-\text{SH}$  group. The transferring mechanism follows:  $-\text{OH}$  of PNIPAM-ATE became  $-\text{O}^-$  in the presence of TEA;  $-\text{O}^-$  attacked carbonyl group and formed intermediate five-membered ring structure; S-C bond broke and produced  $-\text{SH}$  because S is more easily polarized than O. [46] The integral area ratio of  $\text{H}_{\text{C}2}$  and  $\text{H}_{\text{C}1}$  is 10:1, which manifests that the molar ratio of NIPAM unit and MAASE unit in PNIPAM-SH is 10:1, inconsistent with the feed ratio of NIPAM and MAASE for the copolymerization. The incorporation of the  $-\text{SH}$  unit is favor of the formation of nanosilver hydrogel network.

### 3.3. Characterization of the coatings

The amphiphilic coating was firstly examined through UV–Vis spectroscopy to determine the formation of silver nanoparticles (AgNPs) (Fig. 5A). For samples without loading AgOTf during the preparation (SA-0-0 and SA-0-5), there is no characteristic absorption peak for AgNPs at 450–500 nm. In contrast, for SA-1-0 with loading AgOTf and for SA-1-5 with loading AgOTf and hydrophilic polymer PNIPAM-SH, there are peaks at 490 nm and 416 nm, respectively. This demonstrates that the AgNPs were formed during the preparation of the amphiphilic coating. For AgNPs, the shorter wavelength of the absorption peak in UV–Vis spectrum corresponds to smaller particle size. Thus, the introduction of hydrophilic polymer PNIPAM-SH decreases the particle size of AgNPs. This is due to the thiol groups which are in favor of the dispersion of  $\text{Ag}^+$  before forming the AgNPs. [45,47,48] The

morphology and the size of the AgNPs embedded in the amphiphilic coating were examined by TEM. As shown in Fig. 5B, the coating without loading AgOTf and PNIPAM-SH (SA-0-0) does not show any particles or second phases, while the coating with PNIPAM-SH (SA-0-5) shows obvious second phase, which could be attributed to the separation of hydrophilic PNIPAM-SH from hydrophobic bio-based epoxy matrix. The coating loaded with AgOTf (SA-1-0) shows evenly dispersed black spherical particles with size of  $4.8 \pm 1.2$  nm. The crystal space of the particles is 0.281 nm, which is the interplanar spacing of AgNPs (111) space. The formation of AgNPs was further confirmed by XRD (Fig. S1). As can be seen, a family of new peaks at  $38.2^\circ$ ,  $44.3^\circ$ ,  $64.5^\circ$ ,  $77.4^\circ$  for the AgNPs appeared, and these peaks were indexed to (111), (200), (220) and (311). These results are indicative of the successful formation of AgNPs in the amphiphilic coating under UV radiation. For coating loading AgOTf and PNIPAM-SH (SA-1-5), the size of AgNPs is  $2.6 \pm 0.7$  nm smaller than SA-1-0, which is in agreement with the UV–Vis result. Compared with SA-0-5, there is no second phase in SA-1-5, which is due to the formation of nanosilver hydrogel network through the chelation of  $-\text{SH}$  of PNIPAM-SH and AgNPs. The elemental mapping images of SA-1-5 by TEM (Fig. 6) present that elements such as C, O, N, Si and S are homogeneously dispersed in the coating, and Ag is evenly dispersed as nanoparticles, which further demonstrates that the AgNPs and hydrophilic PNIPAM-SH are well penetrated in the bio-based epoxy network.

In order to confirm the formation of IPN in the coatings, the morphology and the glass transition temperature ( $T_g$ ) of the coatings were investigated via SEM and DSC, respectively. As seen from the SEM images (Fig. 7A), there is no particles or second phases in the SA-0-0. The presence of second phases in the SA-0-5 is due to the phase separation of hydrophilic PNIPAM-SH from the hydrophobic bio-based epoxy matrix. This is in agreement with the result from TEM. For SA-1-0, AgNPs cannot be observed from the SEM image, which also indicates that the AgNPs are uniformly dispersed and the particle size is too small to be observed by SEM. From the SEM image of SA-1-5, the

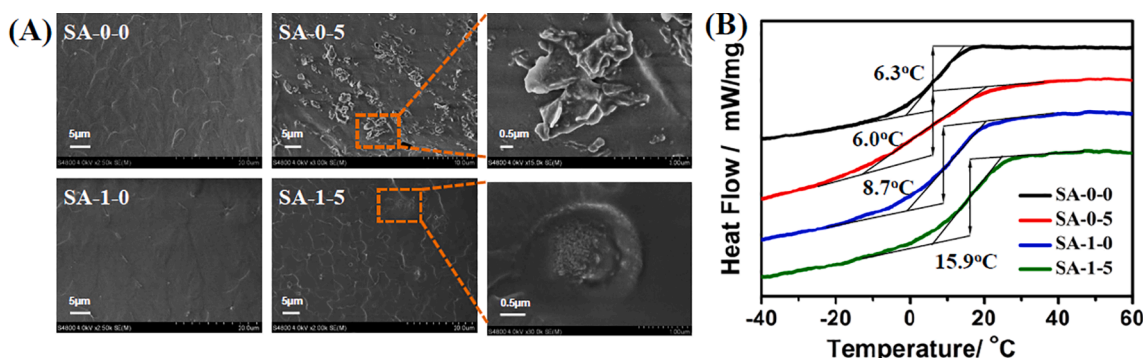


Fig. 7. (A) SEM images and (B) DSC curves of the coatings.

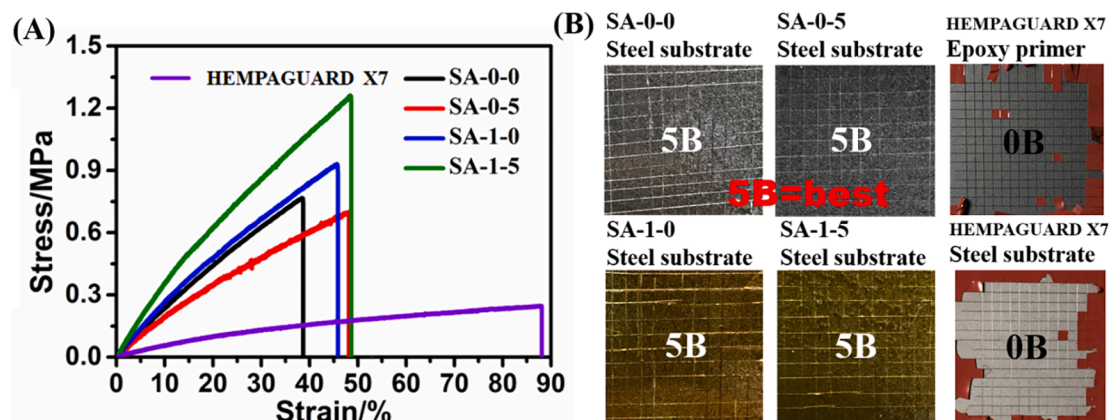


Fig. 8. (A) Representative tensile stress–strain curves of the free films obtained from the coatings; (B) The photographs of the coatings after crosshatch adhesion test.

number of the second phases significantly reduces in comparison to that in SA-0-5, and the interface between the second phases and the main matrix became fuzzy, suggesting that the hydrophilic PNIPAM-SH is well dispersed in the bio-based epoxy network in the presence of AgNPs. Furthermore, the  $T_g$  of the coating is investigated. Fig. 7B presents the DSC curves and the  $T_g$  data of the coatings. Obviously, SA-0-0 and SA-0-5 exhibit similar  $T_g$ , which is due to the fact that the phase separation of hydrophilic PNIPAM-SH reduced its impact on the bio-based epoxy network. SA-1-0 presents a higher  $T_g$  of 8.7 °C than SA-0-0 because AgNPs could block the segmental movement of the bio-based epoxy network. [49–51] Meanwhile, SA-1-5 exhibits greatly increased  $T_g$  value with 15.9 °C in comparison with SA-0-0. This implies that PNIPAM-SH and AgNPs together could prevent the segmental movement of the bio-based epoxy network, which suggests that PNIPAM-SH should be penetrated in the epoxy network. In summary, the AgNPs could cross-link PNIPAM-SH to form hydrogel network, which also have interaction with the –OH in epoxy network. As a result, the hydrogel network is penetrated well in the epoxy network and IPN is formed.

### 3.4. Mechanical properties of the coatings

In order to measure the mechanical properties of the coatings, the free films peeled from the coatings were applied to tensile tests, and the data are shown in Fig. 8. As can be seen in Fig. 8A, the tensile strength and elongation at break of SA-0-0 are 0.69 MPa and 38.7%, respectively. After the introduction of PNIPAM-SH, the elongation at break of SA-0-5 increases by 48.2% comparing to SA-0-0, which might be due to the reason that the PNIPAM-SH increases the flexibility of the coating. While

both tensile strength (0.85 MPa) and elongation at break (45.9%) of SA-1-0 are higher than that of SA-0-0, which demonstrates that AgNPs could simultaneously increase the strength and toughness of the bio-based epoxy coatings. Interestingly, SA-1-5 shows the highest tensile strength (1.23 MPa) and elongation at break (48.6%). The formation of IPN could increase the cross-link density of the coating, which increases mechanical strength. The PNIPAM-SH increases the toughness of the coating, which corresponds to the highest elongation at break. The adhesion is a crucial parameter for coatings, because it can reflect the durability of the coatings. As shown in Fig. 8B, all the bio-based epoxy resin-containing coatings on steel panels give high adhesion of 5B which is the highest adhesion level of this method. While the commercially available PDMS antifouling coating (HEMPAGUARD X7 89900) exhibits poor adhesion on steel substrate even on steel substrate pre-coated with epoxy primer. This might be due to the superior adhesion of bio-based epoxy network toward the steel surface. [52] This result was confirmed by the adhesion strength by drawing experiments. As shown in Fig. S3, the adhesion strength reached 1.86–2.99 MPa higher than the adhesion strengths of the reported PDMS coating (0.4 MPa) [53] and HEMPAGUARD X7 89,900 (almost no adhesion strength toward steel substrate).

### 3.5. Antifouling properties of the coatings

#### 3.5.1. Anti-protein adhesion of the coatings

Anti-protein adhesion is the crucial parameter for the antifouling coatings because the first step of biofouling for the submerged substrate in seawater is the contamination of protein-based biomolecule film. [39,42] Therefore, a representative protein BSA-FITC was utilized to

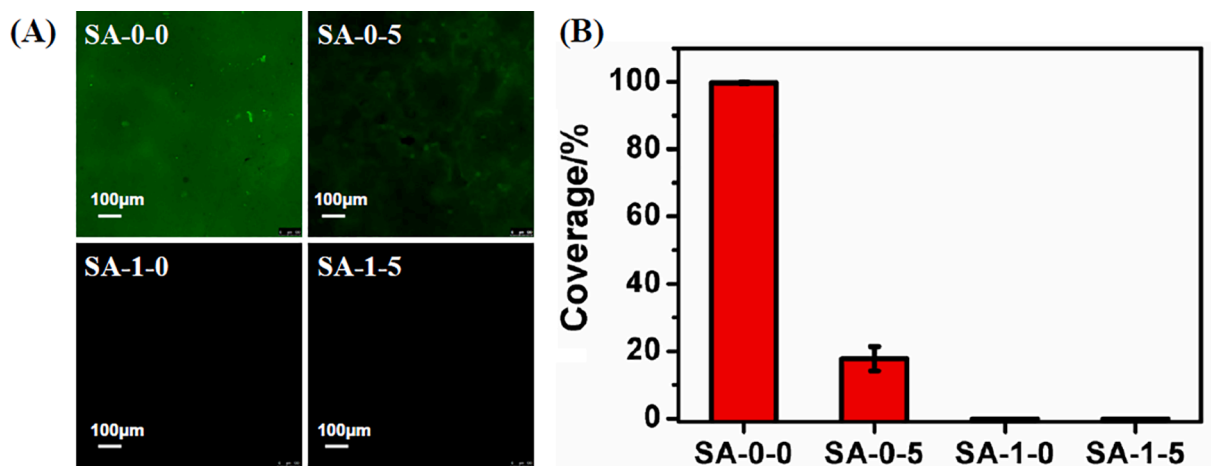


Fig. 9. (A) Fluorescent images of BSA-FITC attachment on the coatings and (B) BSA-FITC coverage area of the coatings.

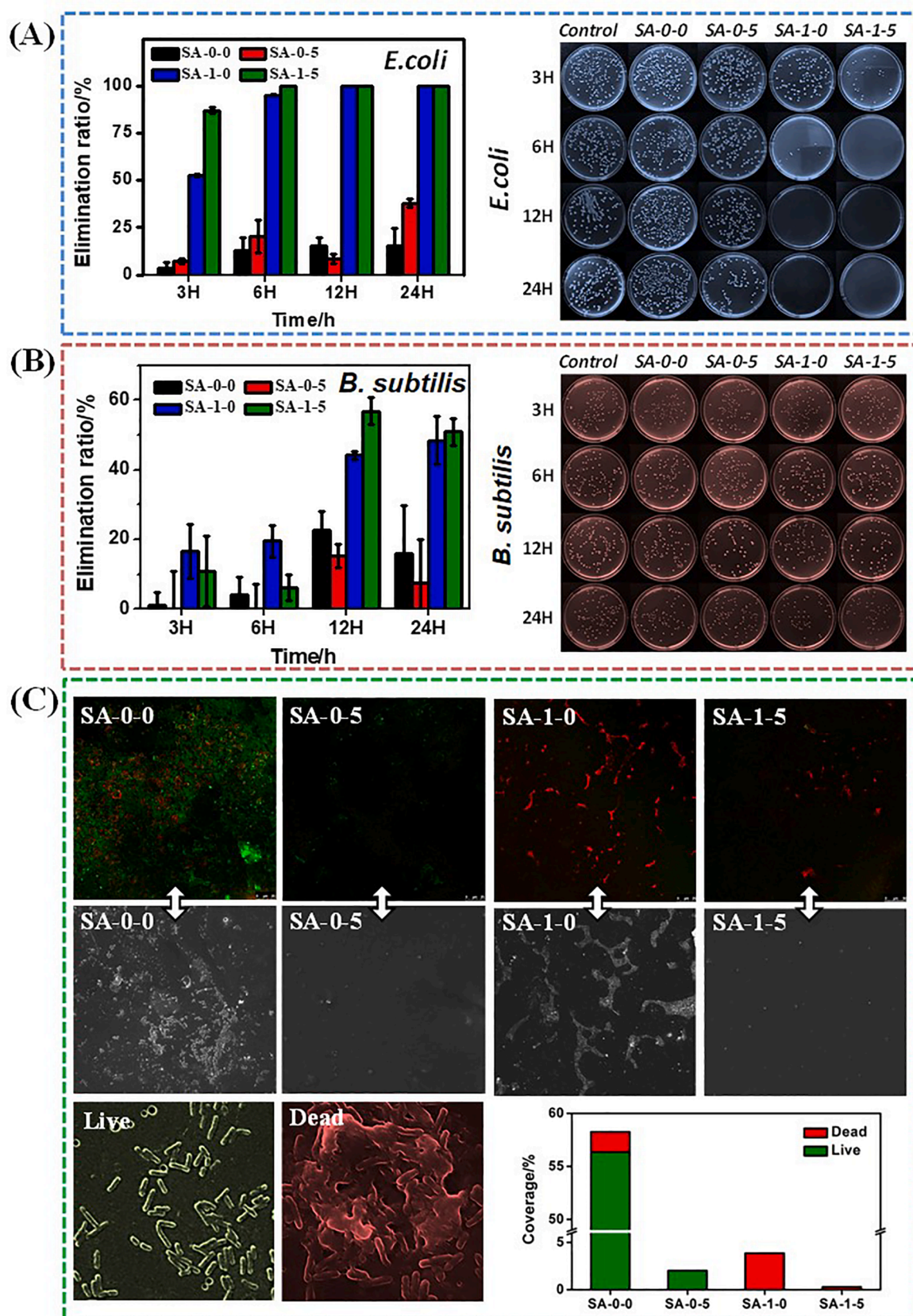


Fig. 10. The killing efficiency of the coatings toward *Escherichia coli* (A) and *Bacillus subtilis* (B) and the related photographs of their colonies incubated on agar plates with the coatings; and (C) the bacteria adhesion of *Escherichia coli* on the coatings.

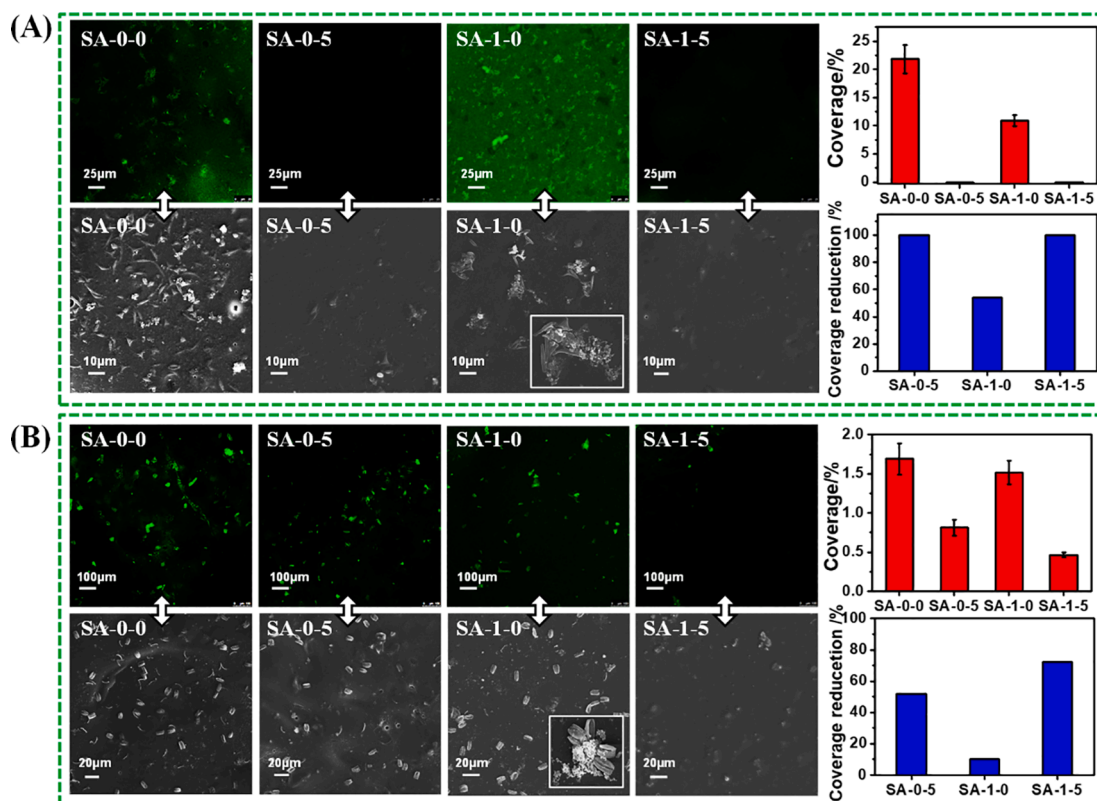


Fig. 11. Algae adhesion of (A) *Phaeodactylum tricornutum* and (B) *Navicula torguatum* on the coatings.

evaluate the anti-protein property of the coatings. Fig. 9 shows that almost no fluorescence signals can be observed on SA-1-0 and SA-1-5, and only 17.7% of the coating surface was attached by BSA-FITCs for SA-0-5, while almost 100% of the coating surface was attached by BSA-FITCs for SA-0-0. These results demonstrate that the hydrophilic PNIPAM-SH can efficiently reduce the protein adsorption on the coatings, and AgNPs can completely prevent the protein attachment, which might be due to the denaturation of protein when exposed to AgNPs. Most proteins have at least one thiol group which could interact with the released  $\text{Ag}^+$  from the AgNPs, resulting in the change of protein's secondary structure and the denaturation of protein.[54] This leads to the better anti-protein adhesion of SA-1-0 than that of SA-0-5.

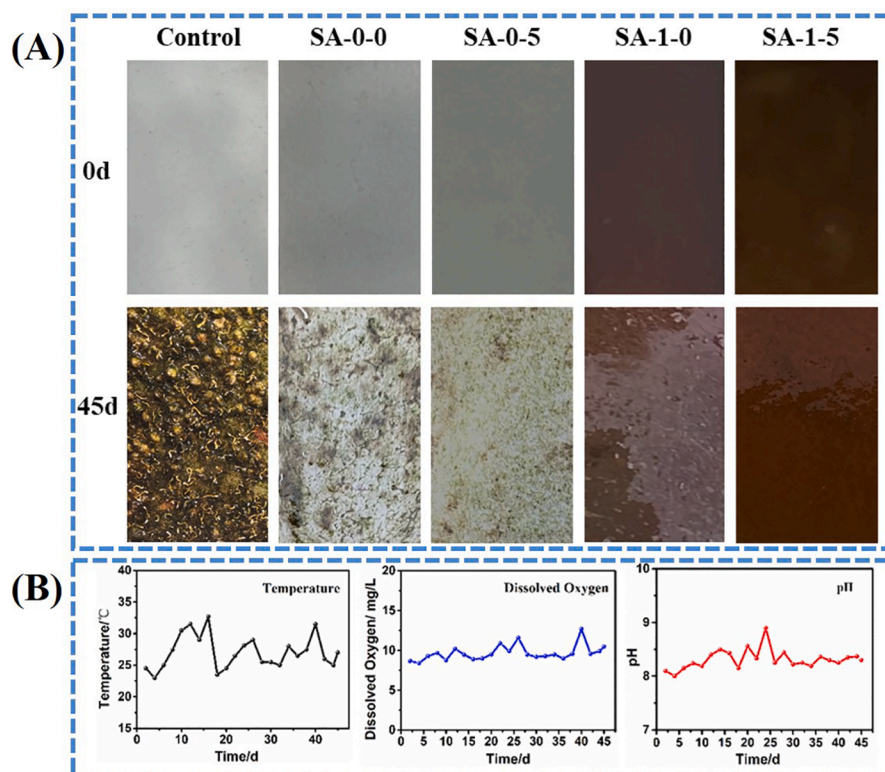
### 3.5.2. Antibacterial properties

Antibacterial adhesion is important for antifouling efficiency. In marine environment, bacteria, as the primary biofouling organism, can be attached on the submerged substrate on the early stage; meanwhile, the attached bacteria can supply abundant nutriment and suitable condition for further settlement of fouling macroorganisms.[1,55] *Escherichia coli*, as one of the common gram-negative bacteria, has been widely used in evaluation of antibacterial adhesion of antifouling coatings[56–58] and *Bacillus subtilis*, as one of the common gram-positive bacteria, generally exists in ocean.[59–60] Therefore, *Escherichia coli* and *Bacillus subtilis* were applied to investigate the antibacterial activity of the coatings through shaking flask method. Fig. 10A and Fig. 10B present the antibacterial results of the coatings toward *Escherichia coli* and *Bacillus subtilis*, respectively. Within 3 h, SA-0-0, SA-0-5, SA-1-0, SA-1-5 could separately kill 3.4%, 7.2%, 52.4% and 87.0% of *Escherichia coli*, and the killing rate increases to 100% after 6 h for SA-1-5 and 12 h for SA-1-0. Although the coatings show weaker antibacterial property toward *Bacillus subtilis*, SA-1-0 and SA-1-5 still exhibit much better antibacterial property than SA-0-0 and SA-0-5. The better antibacterial adhesion toward *Escherichia coli* than that of the *Bacillus subtilis* might be due to the *Bacillus subtilis*' thick cell wall which could be

against cell membrane damage.[61–62] This demonstrates that AgNPs could greatly improve the antibacterial property of the coatings and hydrophilic PNIPAM-SH could also enhance the antibacterial property, especially combined with AgNPs. Fig. 10C shows the fluorescent images of attached *Escherichia coli* on the coatings. SA-0-0 shows a bacteria attachment area of 56.3% (live bacteria) and 1.9% (dead bacteria); SA-0-5 shows a lower bacteria attachment area of 2.0% and all the attached bacteria were live; SA-1-0 also presents low bacteria attachment area of 3.9% and all the attached bacteria were dead; and only 0.3% of the coating surface was attached by dead bacteria for SA-1-5. This suggests that hydrophilic PNIPAM-SH could efficiently prevent the attachment of bacteria on the coatings, and AgNPs could efficiently kill the bacteria, leading to the low attachment area of only dead bacteria. As can be seen in Fig. 10C, the cell membranes of the *Escherichia coli* (killed by AgNPs) are all broken and aggregate together, which suggests that the AgNPs killed the bacteria following the mechanism of damaging the cell membrane and preventing the formation of the cell membrane during the cell division.

### 3.5.3. Alga resistance of the coatings

Alga attachment also occurs at the early stage of marine fouling after the formation of biofilm by protein and bacteria, etc. and provides suitable conditions for other microorganisms. Thus, representative *Phaeodactylum tricornutum* and *Navicula torguatum* were applied to investigate the alga attachment on the surface of the coatings. Fig. 11 presents the alga adhesion results after culturing for 24 h. For *Phaeodactylum tricornutum* (Fig. 11A), SA-0-0, SA-1-0, SA-0-5 and SA-1-5 exhibit alga attachment area of 21.8%, 10.9%, 0%, 0%, respectively. This implies an excellent alga resistance of hydrophilic PNIPAM-SH. It was reported that AgNPs could impact the growth of alga by breaking and degradation of the cell wall.[63] The enhanced fluorescence in the fluorescence image of SA-1-0 should be attributed to the chlorophyll from the killed *Phaeodactylum tricornutum*, which means that AgNPs can kill *Phaeodactylum tricornutum* and improve the alga resistance of the



**Fig. 12.** (A) Images of the coated panels before and after immersion in seawater for 45 days (June 3rd 2020 to July 17th 2020, Ningbo, China). (B) The temperature, oxygen content and pH of seawater during the marine field tests.

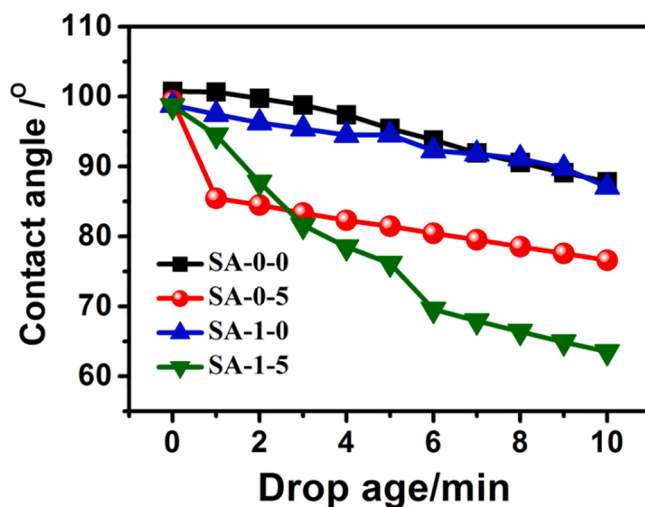
coating. While the killed *Phaeodactylum tricornutum* and the released chlorophyll could be attached on the surface of the coatings, so its effect is not good as SA-0-5 and SA-1-5 which could preventing the attachment of alga from hydrophilic PNIPAM-SH. For *Navicula torguatum* (Fig. 11B), all the coatings show low attachment area from 0.5% to 1.7%, but hydrophilic PNIPAM-SH and AgNPs still reduce the alga resistance of the coatings.

### 3.5.4. Marine field tests

The antifouling performance of the coatings is further evaluated under the real-life marine field at the Meishan Island in Ningbo, China (29°49'N, 122°0'E). Fig. 12A shows the photographs of the coated panels before and after being immersed in seawater for 45 days, and during the test, the temperature, oxygen content and pH of seawater kept stable (Fig. 12B). The surfaces of the all coatings are intact without damage or flaking after 45 days, indicating that the coatings have good mechanical properties and adhesion. Obviously, the uncoated stainless steel panel (the control) was attached a large number of marine organisms including hydroids, barnacles and bryozoans, SA-0-0 was attached a certain amount of hydroids and bryozoans, SA-1-0 was attached a small number of hydroids, and SA-0-5 had no obvious fouling microorganisms but a small amount of alga attached. Encouragingly, SA-1-5 gives the best anti-biofouling properties with almost no marine organisms and only a small amount of sludge attached. These results indicate that hydrophilic PNIPAM-SH and AgNPs together can significantly improve the anti-biofouling performance, and the developed bio-based amphiphilic hydrogel coating shows great potential to protect and prolong the service life of marine materials. The marine field for 5 months was also carried out, the antifouling performance is similar to that reflected by the marine field for 45 days.

### 3.6. Antifouling mechanism

As proved above, the bio-based amphiphilic hydrogel coating SA-1-5



**Fig. 13.** The water dynamic contact angle of the coatings.

shows excellent antifouling properties. Here, the water dynamic contact angles of the coatings were investigated to reveal the antifouling mechanism. Fig. 13 presents the water dynamic contact angles. The change of the contact angles with the test time is utilized to characterize the migration of hydrophilic polymer chains from the inside to the surface of the coating. Obviously, loading only AgNPs does not change the dynamic contact angle obviously, which can be seen from the similar dynamic contact angle curves of SA-0-0 and SA-1-0. While the introduction of hydrophilic PNIPAM-SH significantly increases the reduction rate of the water dynamic contact angle on the coatings. For SA-0-5, the water dynamic contact angle reduces sharply within 1 min, and then keeps the similar dropping rate to that of SA-0-0 and SA-1-0. This suggests that the hydrophilic chain in SA-0-5 can be completely transferred

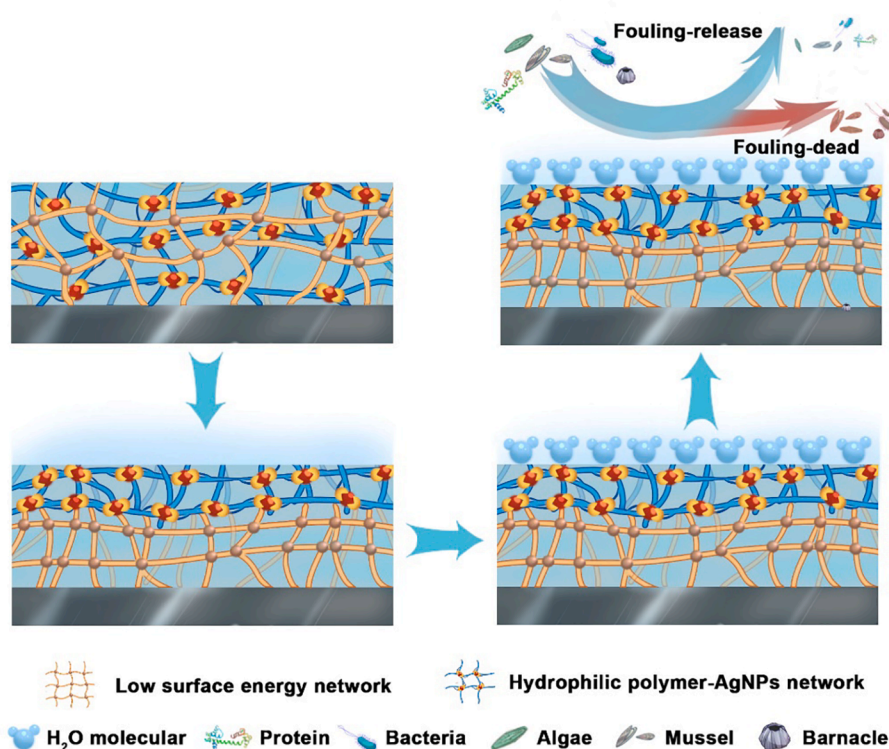


Fig. 14. The supposed antifouling mechanism of the bio-based amphiphilic hydrogel coating.

from the inside to the surface of the coating within 1 min. While for SA-1-5, the water dynamic contact angle drops slower than that of SA-0-5 and keeps high dropping rate until 6 min, and its whole reduction of the contact angle is much higher than SA-0-5. PNIPAM-SH is cross-linked by AgNPs and well dispersed in the bio-based epoxy network, leading to the relatively lower dropping rate of the water dynamic contact angle during the first 1 min and the prolonged migration time compared with SA-0-5, and the transferred hydrophilic chains can also be well dispersed on the surface of the coating. Thus, the antifouling mechanism of bio-based amphiphilic hydrogel coating SA-1-5 can be summarized as Fig. 14. In air, the surface of amphiphilic coating is hydrophobic which can be seen from the high initial contact angle in Fig. 13, which is attributed to the low surface energy of bio-based epoxy network containing hydrophobic siloxane chain. When the amphiphilic coating is immersed in water, hydrophilic PNIPAM-SH can transfer to the surface of the coating and water layer can form to prevent the attachment of organisms such as protein, bacteria, algae, and so on. Meanwhile, with the migration of PNIPAM-SH, AgNPs are also transferred to the surface and effectively kill marine organisms to further improve the antifouling performance.

#### 4. Conclusions

We report the synthesis of bio-based amphiphilic epoxy/nanosilver hydrogel coating with excellent antifouling performance and outstanding mechanical properties. The hydrophobic bio-based epoxy network contributes to the excellent mechanical properties and the highest adhesion (5B), while the hydrophilic hydrogel interpenetrating network provides excellent antifouling properties in the test of both laboratory and marine environment. The superior antifouling performance is attributed to the dynamic migration of hydrophilic compositions including AgNPs to the top layer of the coating to form a thin water film to prevent the attachment of marine organisms as well as kill them (by AgNPs) in the case that attachment occurs. Regarding to the excellent performance as well as bio-based resources, the hydrogel coating

presented in this work shows great potential for marine antifouling industries and is in favor of promoting the development of sustainable antifouling materials.

#### Declaration of Competing Interest

The authors declare that they have no known competing financial interests or personal relationships that could have appeared to influence the work reported in this paper.

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#### Appendix A. Supplementary data

Supplementary data to this article can be found online at <https://doi.org/10.1016/j.cej.2020.128134>.

#### References

- [1] M.S. Selim, M.A. Shenashen, S.A. El-Safty, S.A. Higazy, M.M. Selim, H. Isago, A. Elmarakbi, Recent progress in marine foul-release polymeric nanocomposite coatings, *Prog. Mater. Sci.* 87 (2017) 1–32, <https://doi.org/10.1016/j.pmatsci.2017.02.001>.
- [2] X.i. Chen, S.R. Suwarno, T.H. Chong, D. McDougald, S. Kjelleberg, Y. Cohen, A. G. Fane, S.A. Rice, Dynamics of biofilm formation under different nutrient levels and the effect on biofouling of a reverse osmosis membrane system, *Biofouling* 29 (3) (2013) 319–330, <https://doi.org/10.1080/08927014.2013.772141>.
- [3] J.A. Callow, M.E. Callow, Trends in the development of environmentally friendly fouling-resistant marine coatings, *Nat. Commun.* 2 (1) (2011), <https://doi.org/10.1038/ncomms1251>.
- [4] D.M. Yebra, S. Kiil, K. Dam-Johansen, Antifouling technology—past, present and future steps towards efficient and environmentally friendly antifouling coatings,

- Prog. Org. Coat. 50 (2) (2004) 75–104, <https://doi.org/10.1016/j.porgcoat.2003.06.001>.
- [5] S. Guo, X. Zhu, D. Jańczewski, S.S.C. Lee, T. He, L.S.M. Teo, G.J. Vancso, Measuring protein isoelectric points at the molecular scale by AFM using trace amount of sample, *Nat. Nanotechnol.* 11 (2016) 817–823.
  - [6] S. Guo, S.R. Punireddi, D. Jańczewski, S.S.C. Lee, S.L.M. Teo, T. He, X. Zhu, G. J. Vancso, Barnacle larvae exploring surfaces with variable hydrophilicity: influence of morphology and adhesion of “footprint” proteins by AFM, *ACS Appl. Mater. Interfaces* 6 (16) (2014) 13667–13676, <https://doi.org/10.1021/am503147m>.
  - [7] M.P. Schultz, Effects of coating roughness and biofouling on ship resistance and powering, *Biofouling* 23 (5) (2007) 331–341, <https://doi.org/10.1080/08927010701461974>.
  - [8] M.P. Schultz, J.A. Bendick, E.R. Holm, W.M. Hertel, Economic impact of biofouling on a naval surface ship, *Biofouling* 27 (1) (2011) 87–98, <https://doi.org/10.1080/08927014.2010.542809>.
  - [9] D.J. Blackwood, C.S. Lim, S.L.M. Teo, X. Hu, J. Pang, Macrofouling induced localized corrosion of stainless steel in Singapore seawater, *Corros. Sci.* 129 (2017) 152–160, <https://doi.org/10.1016/j.corsci.2017.10.008>.
  - [10] T.S. Rao, A.J. Kora, P. Chandramohan, B.S. Panigrahi, S.V. Narasimhan, Biofouling and microbial corrosion problem in the thermo-fluid heat exchanger and cooling water system of a nuclear test reactor, *Biofouling* 25 (7) (2009) 581–591, <https://doi.org/10.1080/08927010903016543>.
  - [11] A.G. Nurioglu, A.C.C. Esteves, G. De With, Non-toxic, non-biocide-release antifouling coatings based on molecular structure design for marine applications, *J. Mater. Chem. B* 3 (2015) 6547–6570.
  - [12] E.R. Cronin, A.C. Cheshire, S.M. Clarke, A.J. Melville, An investigation into the composition, biomass and oxygen demand of the fouling community on a tuna aquaculture farm, *Biofouling* 13 (4) (1999) 279–299, <https://doi.org/10.1080/08927019909378386>.
  - [13] J.E. Eckman, D. Thistle, W.C. Burnett, G.L.J. Paterson, C.Y. Robertson, P.J.D. Lambhead, Performance of cages as large animal-exclusion devices in the deep sea, *J. Mar. Res.* 59(2001), 79–95.
  - [14] A.L. Phillippi, N.J. O'Connor, A.F. Lewis, Y.K. Kim, Surface flocking as a possible anti-biofouling, *Aquaculture* 195 (3–4) (2001) 225–238, [https://doi.org/10.1016/S0044-8486\(00\)00556-1](https://doi.org/10.1016/S0044-8486(00)00556-1).
  - [15] Y.i. Liu, X. Suo, Z. Wang, Y. Gong, X. Wang, H. Li, Developing polyimide-copper antifouling coatings with capsule structures for sustainable release of copper, *Mater. Des.* 130 (2017) 285–293, <https://doi.org/10.1016/j.matdes.2017.05.075>.
  - [16] M. Lejars, A. Margaillan, C. Bressy, Fouling release coatings: a nontoxic alternative to biocidal antifouling coatings, *Chem. Rev.* 112 (8) (2012) 4347–4390, <https://doi.org/10.1021/cr200350v>.
  - [17] S. Krishnan, C.J. Weinman, C.K. Ober, Advances in polymers for anti-biofouling surfaces, *J. Mater. Chem.* 18 (29) (2008) 3405, <https://doi.org/10.1039/b801491d>.
  - [18] A. Kaffashi, A. Jannesari, Z. Ranjbar, Silicone fouling-release coatings: effects of the molecular weight of poly(dimethylsiloxane) and tetraethyl orthosilicate on the magnitude of pseudobarnacle adhesion strength, *Biofouling* 28 (7) (2012) 729–741, <https://doi.org/10.1080/08927014.2012.702342>.
  - [19] E.R. Silva, O. Ferreira, P.A. Ramalho, N.F. Azevedo, R. Bayón, A. Igarua, J. C. Bordado, M.J. Calhorda, Eco-friendly non-biocide-release coatings for marine biofouling prevention, *Sci. Total Environ.* 650 (2019) 2499–2511, <https://doi.org/10.1016/j.scitotenv.2018.10.010>.
  - [20] T.H. Duong, J.-F. Briand, A. Margaillan, C. Bressy, Polysiloxane-based block copolymers with marine bacterial anti-adhesion properties, *ACS Appl. Mater. Interfaces* 7 (28) (2015) 15578–15586, <https://doi.org/10.1021/acsami.5b04234>.
  - [21] S.M. Evans, A.C. Birchenough, M.S. Brancato, The TBT ban: out of the frying pan into the fire? *Mar. Pollut. Bull.* 40 (3) (2000) 204–211, [https://doi.org/10.1016/S0025-326X\(99\)00248-9](https://doi.org/10.1016/S0025-326X(99)00248-9).
  - [22] I. Amara, W. Miled, R.B. Slama, N. Ladhari, Antifouling processes and toxicity effects of antifouling paints on marine environment. A review, *Environ. Toxicol. Pharmacol.* 57 (2018) 115–130, <https://doi.org/10.1016/j.etap.2017.12.001>.
  - [23] I. Omae, General aspects of tin-free antifouling paints, *Chem. Rev.* 103 (9) (2003) 3431–3448, <https://doi.org/10.1021/cr030669z>.
  - [24] Z. Lu, Z. Chen, Y.i. Guo, Y. Ju, Y. Liu, R. Feng, C. Xiong, C.K. Ober, L. Dong, Flexible hydrophobic antifouling coating with oriented nanotopography and nonleaking capsaicin, *ACS Appl. Mater. Interfaces* 10 (11) (2018) 9718–9726, <https://doi.org/10.1021/acsami.7b19436.s001>.
  - [25] R.E. Baier, Surface behavior of biomaterials the theta surface for biocompatibility, *J. Mater. Sci. Mater. Med.* 17 (2006) 1057–1062.
  - [26] M. Esfandeh, S.M. Mirabedini, S. Pazokifard, M. Tari, Study of silicone coating adhesion to an epoxy undercoat using silane compounds, *Colloids Surf., A* 302 (1–3) (2007) 11–16, <https://doi.org/10.1016/j.colsurfa.2007.01.031>.
  - [27] M.A.G. Scholten, P.C.W. Thorlakson, A.O. Andersen, A.J. Nielsen, WO/2005/03219, 2005.
  - [28] Q. Xie, J. Pan, C. Ma, G. Zhang, Dynamic surface antifouling: mechanism and systems, *Soft Matter* 15 (6) (2019) 1087–1107, <https://doi.org/10.1039/C8SM01853G>.
  - [29] F. Irani, A. Jannesari, S. Bastani, Effect of fluorination of multiwalled carbon nanotubes (MWCNTs) on the surface properties of fouling-release silicone/MWCNTs coatings, *Prog. Org. Coat.* 76 (2–3) (2013) 375–383, <https://doi.org/10.1016/j.porgcoat.2012.10.023>.
  - [30] L. Cavas, P.G. Yildiz, P. Mimigianni, A. Sapolidis, S. Nitodas, J. Coat. Technol. Res. 15 (2018) 105–120.
  - [31] S.M. Mirabedini, M. Mohseni, S.h. PazokiFard, M. Esfandeh, Effect of TiO<sub>2</sub> on the mechanical and adhesion properties of RTV silicone elastomer coatings, *Colloids Surf., A* 317 (1–3) (2008) 80–86, <https://doi.org/10.1016/j.colsurfa.2007.09.044>.
  - [32] T.P. Galhenage, D. Hoffman, S.D. Silbert, S.J. Stafslien, J. Daniels, T. Miljkovic, J. A. Finlay, S.C. Franco, A.S. Clare, B.T. Nedved, M.G. Hadfield, D.E. Wendt, G. Waltz, L. Brewer, S.L.M. Teo, C.-S. Lim, D.C. Webster, Fouling-release performance of silicone oil-modified siloxane-polyurethane coatings, *ACS Appl. Mater. Interfaces* 8 (42) (2016) 29025–29036, <https://doi.org/10.1021/acsami.6b09484.s001>.
  - [33] S. Amini, S. Kolle, L. Petrone, O. Ahanotu, S. Sunny, C.N. Sutanto, S. Hoon, L. Cohen, J.C. Weaver, J. Aizenberg, N. Vogel, A. Miserez, Preventing mussel adhesion using lubricant-infused materials, *Science* 357 (6352) (2017) 668–673, <https://doi.org/10.1126/science.aai8977>.
  - [34] D.C. Webster, R.J. Pieper, M.J. Nasrullah, WO/2010/042804, 2010.
  - [35] S.K. Rath, J.G. Chavan, S. Sasane, Jagannath, M. Patri, A.B. Samui, B. C. Chakraborty, Two component siloxane modified epoxy foul release coatings: Effect of modulus, surface energy and surface restructuring on pseudobarnacle and macrofouling behavior, *Appl. Surf. Sci.* 256 (8) (2010) 2440–2446, <https://doi.org/10.1016/j.apsusc.2009.10.084>.
  - [36] Z. Chen, B. Chisholm, J. Kim, S. Stafslien, R. Wagner, S. Patel, J. Daniels, L.V. Wal, J. Li, K. Ward, M. Callow, S. Thompson, C. Siripiprom, UV-curable, oxetane-toughened epoxy-siloxane coatings for marine fouling-release coating applications, *Polym. Int.* 57 (6) (2008) 879–886, <https://doi.org/10.1002/pi.2422>.
  - [37] A. Sokolova, N. Cilz, J. Daniels, S.J. Stafslien, L.H. Brewer, D.E. Wendt, F.V. Bright, M.R. Detty, A comparison of the antifouling/foul-release characteristics of non-biocidal xerogel and commercial coatings toward micro- and macrofouling organisms, *Biofouling* 28 (5) (2012) 511–523, <https://doi.org/10.1080/08927014.2012.690197>.
  - [38] P.J. Molino, S. Childs, M.R. Eason Hubbard, J.M. Carey, M.A. Burgman, R. Wetherbee, Development of the primary bacterial microfouling layer on antifouling and fouling release coatings in temperate and tropical environments in Eastern Australia, *Biofouling* 25 (2) (2009) 149–162, <https://doi.org/10.1080/08927010802592917>.
  - [39] H. Guo, J. Yang, W. Zhao, T. Xu, C. Lin, J. Zhang, L. Zhang, Direct formation of amphiphilic crosslinked networks based on PVP as a marine anti-biofouling coating, *Chem. Eng. J.* 374 (2019) 1353–1363, <https://doi.org/10.1016/j.cej.2019.06.025>.
  - [40] Z. Cao, S. Jiang, Super-hydrophilic zwitterionic poly(carboxybetaine) and amphiphilic non-ionic poly(ethylene glycol) for stealth nanoparticles, *Nano Today* 7 (5) (2012) 404–413, <https://doi.org/10.1016/j.nantod.2012.08.001>.
  - [41] H. Wang, Y. Hu, D. Lynch, M. Young, S. Li, H. Cong, F. Xu, G. Cheng, Zwitterionic polyurethanes with tunable surface and bulk properties, *ACS Appl. Mater. Interfaces* 10 (43) (2018) 37609–37617, <https://doi.org/10.1021/acsami.8b10450.s001>.
  - [42] W. Zhao, J. Yang, H. Guo, T. Xu, Q. Li, C. Wen, X. Sui, C. Lin, J. Zhang, L. Zhang, Slime-resistant marine anti-biofouling coating with PVP-based copolymer in PDMS matrix, *Chem. Eng. Sci.* 207 (2019) 790–798, <https://doi.org/10.1016/j.ces.2019.06.042>.
  - [43] S. Guo, R. Quintana, M. Cirelli, D.Z.S. Toa, V.A. Vasantha, E.S. Kooij, D. Jańczewski, G.J. Vancso, Brush swelling and attachment strength of barnacle adhesion protein on zwitterionic polymer films as a function of macromolecular structure, *Langmuir* 35 (24) (2019) 8085–8094, <https://doi.org/10.1021/acs.langmuir.9b00918.s001>.
  - [44] X. Zhu, S. Guo, T. He, S. Jiang, D. Jańczewski, G.J. Vancso, Engineered, robust polyelectrolyte multilayers by precise control of surface potential for designer protein, cell, and bacteria adsorption, *Langmuir* 32 (5) (2016) 1338–1346, <https://doi.org/10.1021/acs.langmuir.5b04118>.
  - [45] S. Tian, D. Jiang, J. Pu, X. Sun, Z. Li, B. Wu, W. Zheng, W. Liu, Z. Liu, A new hybrid silicone-based antifouling coating with nanocomposite hydrogel for durable antifouling properties, *Chem. Eng. J.* 370 (2019) 1–9, <https://doi.org/10.1016/j.cej.2019.03.185>.
  - [46] G. Wang, L. Peng, Y. Zheng, Y. Gao, X. Wu, T. Ren, C. Gao, J. Han, Novel triethylamine catalyzed S → O acetyl migration reaction to generate candidate thiols for construction of topological and functional sulfur-containing polymers, *RSC Adv.* 5 (8) (2015) 5674–5679, <https://doi.org/10.1039/C4RA09842K>.
  - [47] D. Jiang, Q. Xue, Z. Liu, J. Han, X. Wu, A novel anti-algal nanocomposite hydrogel based on thiol/acetyl thioester groups chelating with silver nanoparticles, *New J. Chem.* 41 (2017) 271–277.
  - [48] D. Jiang, G. Wang, F. Zheng, J. Han, X. Wu, Novel thermo-sensitive hydrogels containing polythioether dendrons: facile tuning of LCSTs, strong absorption of Ag ions, and embedment of smaller Ag nanocrystals, *Polym. Chem.* 6 (4) (2015) 625–632, <https://doi.org/10.1039/C4PY01158A>.
  - [49] Y. Yagci, M. Sangermano, G. Rizza, A visible light photochemical route to silver-epoxy nanocomposites by simultaneous polymerization–reduction approach, *Polymer* 49 (24) (2008) 5195–5198, <https://doi.org/10.1016/j.polymer.2008.09.068>.
  - [50] M.S. Santhosh, K. Natarajan, Antibiofilm activity of epoxy/Ag-TiO<sub>2</sub> polymer nanocomposite coatings against *Staphylococcus aureus* and *Escherichia coli*, *Coatings* 5(2015), 95–114.
  - [51] M. Cobos, I. De-La-Pinta, G. Quindós, M.J. Fernández, M.D. Fernández, One-step eco-friendly synthesized silver-graphene oxide/poly(vinylalcohol) antibacterial nanocomposite, *Carbon* 150 (2019) 101–116.
  - [52] S. Ma, D.C. Webster, F. Jabeen, Hard and flexible, degradable thermosets form renewable bioresources with the assistance of water and ethanol, *Macromolecules* 49 (2016) 3780–3788.

- [53] P. Hu, Q. Xie, C. Ma, G. Zhang, Fouling resistant silicone coating with self-healing induced by metal coordination, *Chem. Eng. J.* 406 (2021) 126870–126878.
- [54] H.S. Jiang, Y.Z. Zhang, Z.W. Lu, R. Lebrun, B. Gontero, W. Li, Interaction between silver nanoparticles and two dehydrogenases: role of thiol groups, *Small* 15 (27) (2019) 1900860, <https://doi.org/10.1002/smll.201900860>.
- [55] M. Salta, J.A. Wharton, Y. Blache, K.R. Stokes, J.-F. Briand, Marine biofilms on artificial surfaces: structure and dynamics, *Environ. Microbiol.* 15 (2013) 2879–2893.
- [56] L. Zhang, J. Sha, R. Chen, Q. Liu, J. Liu, J. Yu, H. Zhang, C. Lin, W. Zhou, J. Wang, Surface plasma Ag-decorated Bi<sub>5</sub>O<sub>7</sub>I microspheres uniformly distributed on a zwitterionic fluorinated polymer with superfunctional antifouling property, *Appl. Catal. B* 271 (2020) 118920–118928, <https://doi.org/10.1016/j.apcatb.2020.118920>.
- [57] L. Zhang, J. Sha, R. Chen, Q. Liu, J. Liu, J. Yu, H. Zhang, C. Lin, J. Wang, Three-dimensional flower-like shaped Bi<sub>5</sub>O<sub>7</sub>I particles incorporation zwitterionic fSha, R. Chen, Q. Liu, J. Liu, J. Yu, H. Zhang, C. Lin, J. Wang, Three-dimensional flower-like shaped Bi<sub>5</sub>O<sub>7</sub>I particles incorporation zwitterionic 88.t.
- [58] L. Zhao, R. Chen, L. Lou, X. Jing, Q. Liu, Jingyuan Liu, Jing Yu, Peili Liu, Jun Wang, Layer-by-layer-assembled antifouling films with surface microtopography inspired by *Laminaria japonica*, *Appl. Surf. Sci.* 511 (2020) 145564, <https://doi.org/10.1016/j.apsusc.2020.145564>.
- [59] U. Eduok, M. Khaled, A. Khalil, R. Suleiman, B.E. Ali, Probing the corrosion inhibiting role of thermophilic *Bacillus licheniformis* biofilm on steel in a saline axenic culture, *RSC Adv.* 6 (2016) 18246–18256.
- [60] N. Guo, Y. Wang, X. Hui, Q. Zhao, Z. Zeng, S. Pan, Z. Guo, Y. Yin, T. Liu, Marine bacteria inhibit corrosion of steel via synergistic biomineralization, *J. Mater. Sci. Technol.* 66 (2021) 82–90, <https://doi.org/10.1016/j.jmst.2020.03.089>.
- [61] A. Le, L.T. Tam, P.D. Tam, P.T. Huy, T.Q. Huy, N.V. Hieu, A.A. Kudrinskiy, Yu A. Krutyakov, Synthesis of oleic acid-stabilized silver nanoparticles and analysis of their antibacterial activity, *Mater. Sci. Eng., C* 30 (6) (2010) 910–916, <https://doi.org/10.1016/j.msec.2010.04.009>.
- [62] S.W. Kim, Y.W. Baek, Y.J. An, Assay-dependent effect of silver nanoparticles to *Escherichia coli* and *Bacillus subtilis*, *Appl. Microbiol. Biotechnol.* 92 (5) (2011) 1045–1052, <https://doi.org/10.1007/s00253-011-3611-x>.
- [63] A. Dash, A.P. Singh, B.R. Chaudhary, S.K. Singh, D. Dash, Effect of silver nanoparticles on growth of eukaryotic green algae, *Nano-Micro Lett.* 4 (3) (2012) 158–165, <https://doi.org/10.1007/BF03353707>.